

Refreshable self-polishing superhydrophobic coating on Mg alloy to prohibit corrosion and biofouling in marine environment

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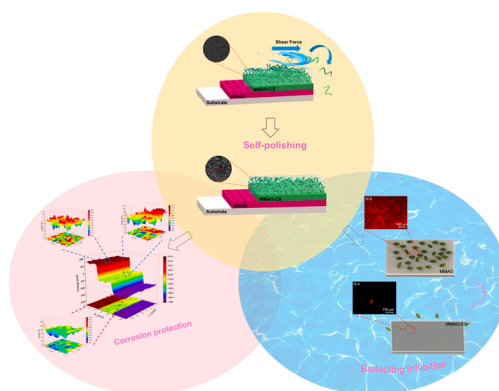
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HIGHLIGHT

- A refreshable super-hydrophobic coating is bound onto AZ31.
- Superhydrophobic candle soot demonstrates self-polishing.
- Self-polishing carbon soot highly inhibits corrosion and biofouling.

GRAPHICAL ABSTRACT



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ABSTRACT

Corrosion susceptibility is a critical obstacle for Mg alloy usage in harsh environments. Bioinspired superhydrophobic surface (SHS) affords a new way to combat corrosion. However, problems, e.g., water vapor condensation and air dissolution, ceaselessly hurdle practical usage of SHS for long-term corrosion inhibition. In this report, targeting the corrosion inhibition of Mg alloy in seawater, a refreshable self-polishing superhydrophobic coating is bound onto AZ31 Mg alloy bridging via a ceramic/paraffin buffer layer to solve the two problems encountered by traditional SHS. Firstly, the liquid paraffin is injected into the porous ceramic structure constructed by micro-arc oxidation (MAO). Then, carbon soot (CS) is grown onto the ceramic/paraffin layer via a combustion method. The internal CS layer is firmly bonded due to the molten paraffin. Meanwhile, the external superhydrophobic CS layer is soft and floppy to perform self-polishing property under dynamic water flow condition. The refreshable feature endows the regeneration of SHS. Using water jet flush (5.8 m/s) and immersion test up to 168 h, Mg alloy covered ceramic layer and carbon soot (MMAO-CS) coating demonstrates excellent stability. Using diatoms as target organisms to illustrate seawater anti-biofouling property, a few diatoms are found on MMAO-CS even after immersion for 10 d due to the low modulus and self-polishing of the external superhydrophobic layer invaded by the water phase and bio-organisms. For prohibiting corrosion in

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seawater environment, MMAO-CS coating possesses prominent resistance with $|Z|_{0.01 \text{ Hz}}$ ($2.82 \times 10^7 \Omega \text{ cm}^2$) of 6 orders of magnitude larger than bare AZ31 ($42.9 \Omega \text{ cm}^2$), enabling its potential usage in marine environment.

1. Introduction

Magnesium alloy, as an outstanding engineering metal material, is widely used in aerospace, vehicles and other fields because of its superiority in mechanical performance, e.g., light weight, high toughness and stiffness [1]. However, the standard electrode potential of the main component magnesium in the alloy is as low as -2.36 V , and the as-formed oxide film in natural atmosphere is unable to act as a rigid barrier to counter the corrosion from the environment. Thus, the corrosion resistance is a critical problem for the usage of Mg alloy [2]. At present, micro-alloying and surface treatment are the main routes to improve the corrosion resistance of magnesium alloys [3]. However, based on microalloying, the improvement of the intrinsic corrosion resistance of magnesium alloy is still very limited, and structural components fabricated by the alloy are difficult for direct utilization in certain environment [4]. The surface treatment to enhance corrosion resistance of magnesium alloy has aroused wide interest in recent decades [5]. Nowadays, researchers have developed various treating routes to reduce the corrosion rate of magnesium alloy, including anodic oxidation, chemical conversion, micro-arc oxidation (MAO), electroplating, electroless plating, thermal spraying and laser surface processing [5–9]. Among all these surface treatment methods for Mg alloy protection, MAO technology is very promising because of its simple process, low environmental pollution, excellent bond strength with substrate, relatively high thickness and tough hardness on the Mg alloys [10–12].

The ceramic oxide layer formed by MAO can block the direct contact between magnesium alloy substrate and the corrosion medium to a certain extension, so that the corrosion resistance of the metal will be improved [13,14]. However, the micro-porous structure defects caused by plasma discharge provide a transportation channel for the corrosive medium [15], giving the chance for the corrosive species to reach the substrate and corrode the magnesium alloy. Therefore, only a porous ceramic layer formed by MAO is not an effective and long-term barrier to protect Mg alloy.

Enlightened by the extreme water repellency phenomenon on the surface of particular animals and plants, superhydrophobic surface (SHS) has been widely studied in the field of metal corrosion protection [16–18]. Coating based on SHS uses a stored air cushion to block the direct contact between aqueous corrosive solution and the substrate surface, so as to protect the substrate from corrosion medium [19,20]. However, there are still some drawbacks to hurdle the longevity and the duration of SHS in the seawater or marine atmosphere conditions. One is the capillary condensation effect from the micro/nano-space existing in the SHS matrix. During immersion in seawater phase or maintaining in the humid atmospheric conditions, the water molecule will condense in the channel of the matrix, so that the extreme hydrophobicity will loose with the coverage of the water layer [21]. The other aspect for shortening the longevity of SHS is the exhaustion of entrapped air in the matrix. During immersion in water phase, the as-trapped air will spontaneously dissolve into the water phase [22], and the SHS phenomenon will diminish and finally die out. Up to now, these problems are still remaining unsolved and regarded to induce the impasse for practical usage of SHS.

For Mg alloy, in previous reports, SHS has been constructed onto the

substrate surface for corrosion inhibition [23–29]. However, the direct anchoring of SHS seems not a good idea for protecting the underneath Mg alloy [30]. Due to the high activity, Mg alloy will directly react with the coming water phase introduced by capillary condensation or dissolution of air cushion to form the hydrophilic corrosion product, leading to the failure of SHS. Therefore, introducing a versatile buffer layer to isolate SHS and the underneath Mg alloy is a brilliant idea to elongate the corrosion inhibition. When superhydrophobic coating is constructed onto the ceramic buffer obtained via MAO, the corrosion resistance of magnesium alloy is further improved [1,10,31], since the achieved porous structure via MAO can enhance the bonding strength of superhydrophobic coating by interlocking effect. In recent years, superhydrophobic coatings based on MAO buffer layer are reported [11, 32–37]. Most of them use further treatment, such as hydrothermal method, dip coating and deposition method, to reduce the holes formed in MAO ceramic coating. To our best knowledge, using both paraffin and carbon soot to seal MAO holes has not been reported.

On the other hand, the carbon soot as the nanoscale material can be easily obtained during the insufficient combustion of the fuels without needing complicated and particular facilities [38]. An important characteristic of carbon soot particles is their intrinsic hydrophobicity, which can enable direct fabrication of highly water-repellent SHS without further post-functionalization steps [39]. The interaction of CS deposition layer is relatively weak, and the coating is easy to peel off upon mechanical effect. However, from the philosophy perspective, a weakness is not always a weakness. The low binding force and weak mechanical stability of the candle soot-based superhydrophobic material will find certain practical usage, such as the inhibition of the biofouling and so forth.

In this study, a super-hydrophobic layer based on composite coating composed of carbon soot and ceramic oxide via MAO is used for corrosion protection of magnesium alloy. Using a facile carbon soot deposition method by combustion of candle, the super-hydrophobic layer is formed. By infusing paraffin into the porous ceramic oxide layer, the inner carbon soot layer will firmly bond to the solid substrate, and the adhesion of soot coating is effectively improved. Meanwhile, the external layer will be relatively loose to enable an aerosol characteristic, which will perform a self-polishing feature during serving in seawater and atmospheric conditions. In dynamic water condition, the SHS will self-polish to lose the very external soot, and the remaining underneath matrix still remains the extreme wettability. In atmosphere condition, when the salt fog resides on SHS surface, the flush water, e.g., rainfall, will clean the aerosol with the contamination, so that the corrosion will be steadily prohibited. Moreover, due to the facile preparation process, the coating can realize revival by in situ preparation, especially for the usage in atmospheric and periodic immersion scenario. This affords a new way to combat the two problems tangling the practical usage of SHS.

2. Materials and methods

2.1. Materials

The composition of the commercial AZ31 is shown in Table 1. AZ31 ingot was cut into plate with the dimension as $20 \text{ mm} \times 20 \text{ mm} \times 5 \text{ mm}$, and the surface was ground down to 2000 grade using sandpaper. After being degreased by using distilled water and ethanol, the AZ31 plate was placed in a thermal oven having modest temporary. The experimental materials including candle, green tea, milk, juice and blue ink were purchased from the local supermarket. Sodium hydroxide (NaOH) and hydrochloric acid (HCl) as the representative alkali and acid were both

Table 1
Elemental composition of AZ31 alloy (wt%).

Element	Al	Zn	Mn	Mg
wt%	2.50–3.00	0.70–1.30	> 0.20	Balance

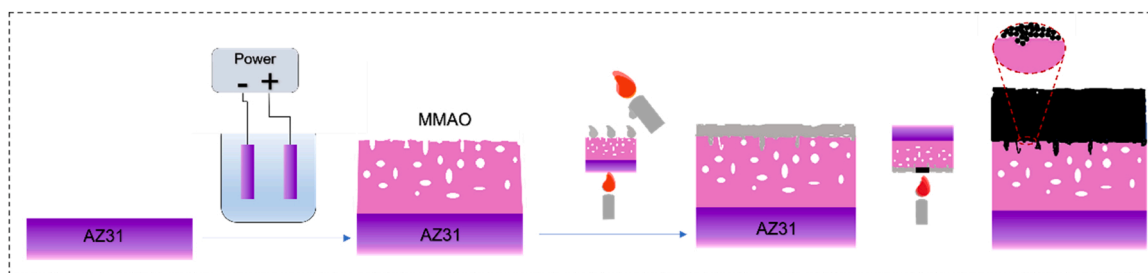


Fig. 1. Schematic illustration of the route for preparing superhydrophobic MMAO-candle soot composite coating on AZ31 Mg alloy.

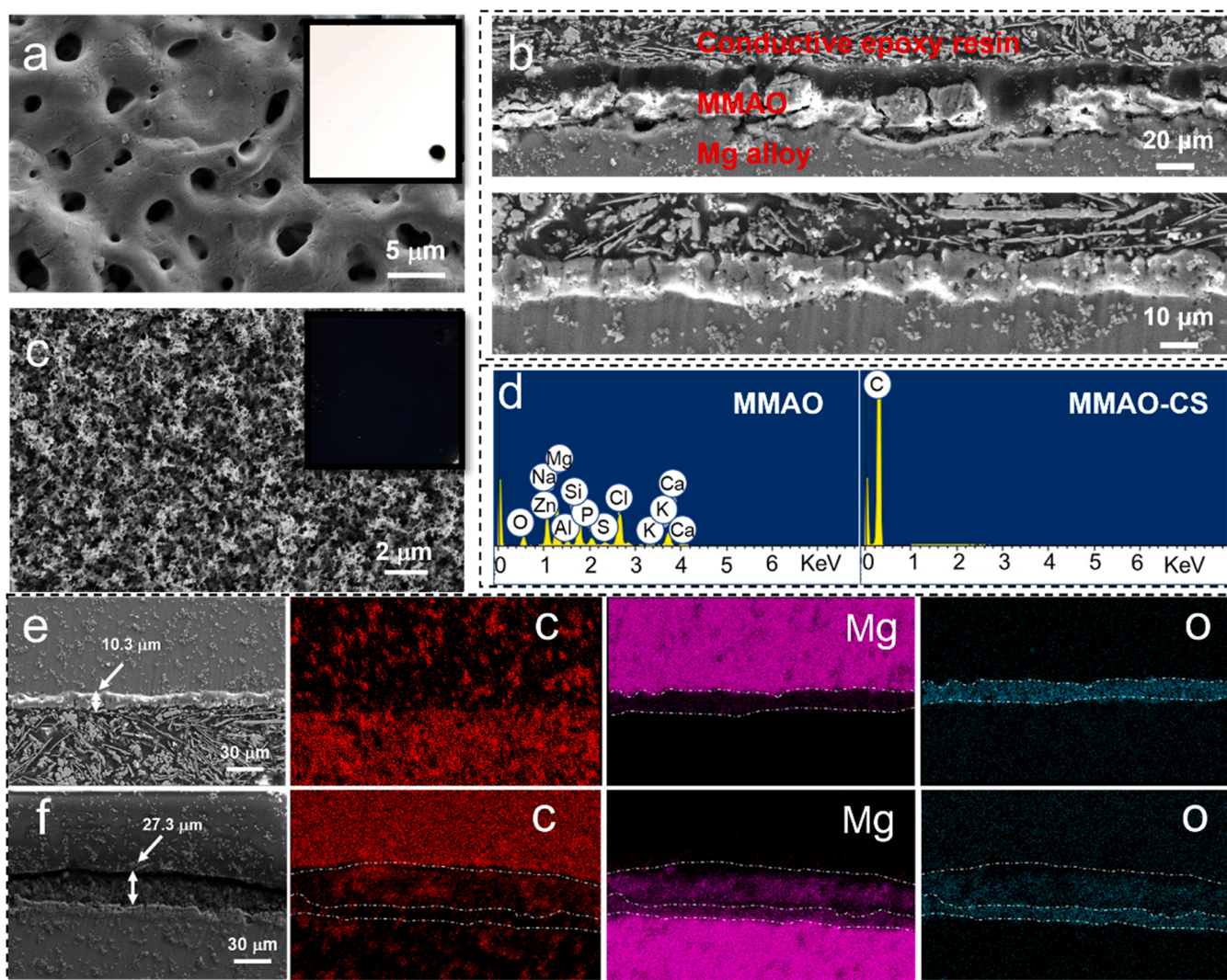


Fig. 2. Morphology of MMAO and MMAO-CS on Mg alloy substrate. (a) SEM of MMAO coating, and the inset is the optical image of MMAO plate with a hole. (b) Cross-sectional SEM morphology of MMAO. (c) SEM of MMAO-CS coating. Inset is the optical image of the as-prepared MMAO-CS. (d) EDS of both MMAO and MMAO-CS. (e and f) Cross-sectional SEM morphology and element distribution of MMAO coating and MMAO-CS coating.

purchased from Aladdin without further treatment before usage.

2.2. Preparation of MMAO and MMAO-CS composite coatings and characterization

The pretreated AZ31 substrate was anodized by using a commercially available micro-arc oxidation (MAO) equipment. The coating on Mg alloy via MAO is denoted as MMAO, and it was prepared at 300 V at 500 Hz with a procedure of 3 min by employing an AC power supplier.

The electrolyte contained in the solution were NaOH, Na_2SiO_3 and KF. After the preparation, the as-fabricated oxidized sample was washed by deionized water to get rid of the remaining electrolyte as the impurity. The detailed preparation process of the MMAO-candle soot composite coating is shown in Fig. 1. To get MMAO-CS composite coating, a candle was lit to evenly drop the molten paraffin on the MMAO coating surface. MMAO coating was heated to infuse the paraffin into the porous matrix. Then, it was gently shaken to ensure the even coverage of the entire coating surface. Next, a distance of approximately 2–3 cm was

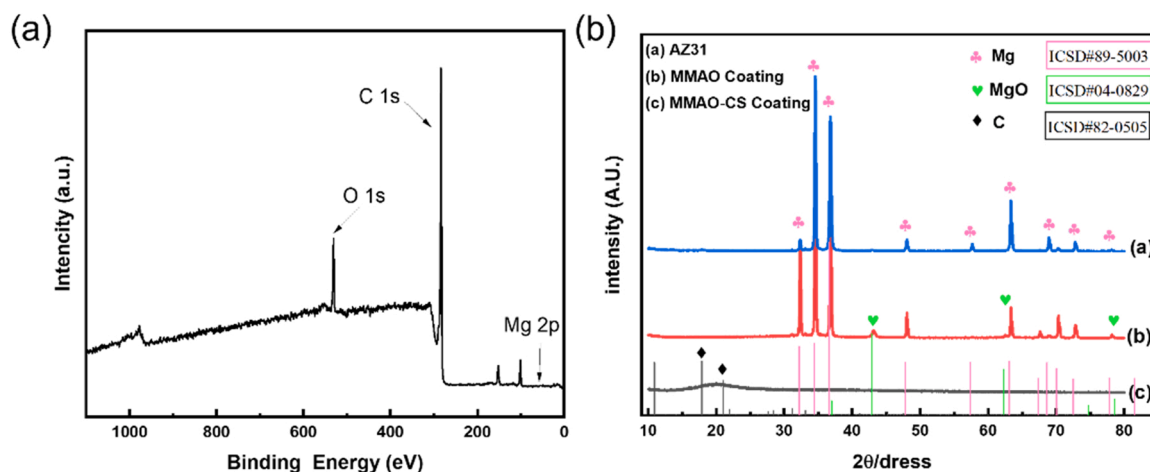


Fig. 3. Element and phase composition of the as-fabricated coatings. (a) XPS of the MMAO-CS coating. (b) XRD patterns of the bare AZ31 substrate, MMAO and MMAO-CS coating.

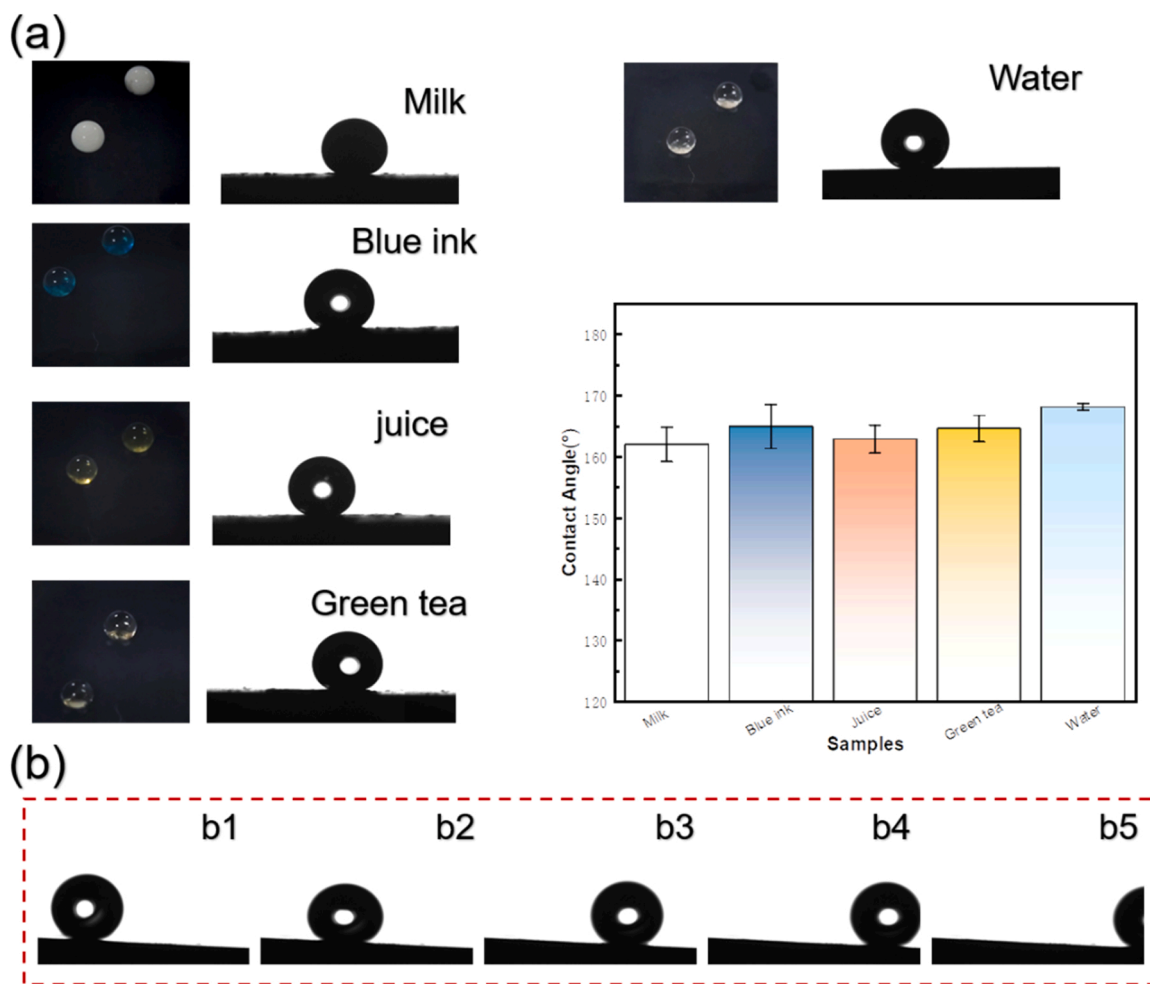


Fig. 4. Wettability characterization of MMAO-CS. (a) Contact angle of milk, blue ink, juice, green tea and water droplet on the surface of MMAO-CS coating. (b) Rolling scenario of water droplet on the surface of MMAO-CS.

maintained between the candle flame and the sample of MMAO covered by paraffin to accommodate the carbon soot. The MMAO-CS coating was uniformly coated with a CS layer by the flame. The methods for knowing surface morphology, chemical composition characterization, wetting behavior and anti-corrosion resistance of coating are included in

Supporting Information. The adhesion strength of MMAO CS-AZ31 and MMAO-AZ31 interfaces was respectively measured using a Micro Scratch Tester (Revetest, CSM, Switzerland). The scratch probe was diamond texture with radius of 200 μm . The scratch test was carried out at a constant scratch rate of 5.05 mm/min. During the scratching, the

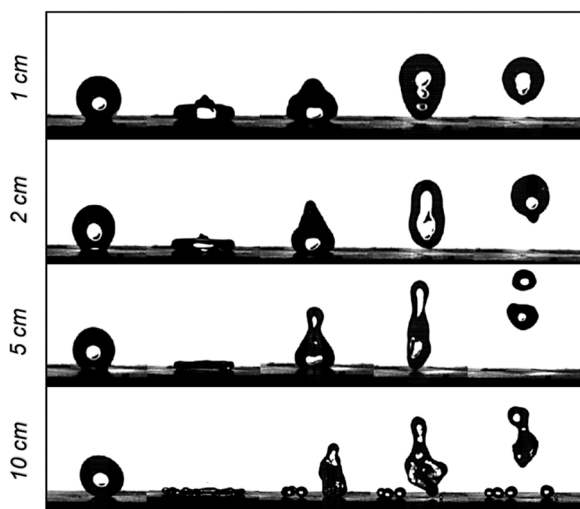


Fig. 5. Selected snapshots of dynamic droplet-droplet interaction between water droplet released from different height with MMAO-CS.

load linearly increased to 100 N with a loading rate of 10 N/min. The measurement was continuously recorded until the scratch length reached 5 mm.

2.3. Biofouling inhibition of Mg alloy based on different coatings

The bio-adhesion behavior of MMAO and MMAO-CS coatings was investigated, and diatoms algae were used as the probe organisms. The process of collection and cultivation of diatoms was in accordance with our previous reports [40,41]. To achieve the average bio-adhesion state from the statistical perspective, three parallel samples were prepared. Enduring immersion for 1, 5 and 10 d, the soaked samples including MMAO and MMAO-CS were brought out of the six-hole plate for evaluating the biofouling state.

3. Results and discussion

3.1. Coating preparation, morphology and composition

Following the schematic diagram shown in Fig. 1, MMAO and MMAO-CS coatings are constructed onto Mg alloy substrate. Fig. 2 has given the surface characterization of the sample from different aspects. For top view of the MMAO coating (Fig. 2a), porous and uneven structure of the coating can be observed, which is attributed to the rapid oxidation process afforded by the micro-arc plasma discharge. Meanwhile, the cross-sectional SEM morphology of MMAO is shown in Fig. 2b, which indicates that the thickness of MMAO coating is ca. 10 μm . The channel in the porous oxide layer can be clearly distinguished due to the severe oxidation process afforded by MAO process. Generally, MMAO contains two sub-layers, a porous outer layer and a dense inner layer. From this indication of cross-sectional SEM, it is assured that only MMAO coating cannot effectively afford the barrier to inhibit the corrosion. For the coating on the metal to be protected, the defects are highly requested to avoid. The pores in MMAO are concerned as the plentiful defects intrinsically born in the coating. During the practical service, the defects act as the vulnerable sites for penetrating the aggressive corrosion species. Therefore, the corrosion inhibition from the primitive MMAO coating will not last long period in the harsh environment, e.g., marine environment. The further enhancement is indispensable for promoting the corrosion inhibition effect of the coating. As shown in Fig. 2c, after the candle soot (CS) deposition to realize the MMAO-CS coating, a uniformly porous structure is formed onto MMAO at the external surface. For the deposited nanometer scale CS, it is mainly filament structure. The chemical composition

comparison is shown as Fig. 2d. Before CS deposition, the containing elements on the MMAO surface are Mg, Al, O and so forth. The complex elemental composition is due to the MAO process, which involves various salts to participate the oxidation. After the CS deposition, all the aforementioned elements on MMAO have been screened, and the carbon element has become the single detectable element, indicating the entire coverage of the carbon soot on MMAO. Using EDS mapping technology can reveal the layer structure of the crossing section from the element distribution perspective. As shown in Fig. 2e, the main elements of MMAO in the exposed crossing section are shown to be Mg and O, which correspond to the existence of MgO. In Fig. 2f, MMAO-CS coating is 27.3 μm , which is thicker than that of MMAO coating. It is seen that the boundary between the MMAO-CS coating and MMAO is ambiguous, suggesting that part of CS layer is sealed into the pore of MMAO matrix.

Further, XPS is employed to obtain more composition information of MMAO-CS. As observed in Fig. 3a, the element peaks of C, O, and Mg appear on the surface of MMAO-CS coating. The XRD pattern of the bare Mg alloy substrate show that it mainly consists of Mg matrix as the substrate (Fig. 3b). For MMAO, the characteristic peak located at 43° is ascribed to MgO phase, implying the successful fabrication of the ceramic coating on Mg alloy AZ31 by MAO process. Also, the characteristic peaks of α -Mg as the substrate are revealed, indicating the relatively thin layer on the substrate. For MMAO-CS, a bump locating at $16\text{--}25^\circ$ is observed in the pattern, implying the surface of coating is composed of carbon black in the amorphous state. The peaks corresponding to Mg and MgO are absent in XRD pattern of the MMAO-CS coating. The main reason is that the CS layer is very thick. After penetration this layer, X-ray attenuates as the very weak radiation, which cannot further reflect the structural information of underneath MMAO and Mg substrate.

In addition, Raman spectra of AZ31, MMAO and MMAO-CS are shown in Fig. S1. No obvious peak is observed in the Raman spectra of AZ31 and MMAO. Meanwhile, there are two Raman peaks at 1350 cm^{-1} and 1587 cm^{-1} in MMAO-CS coating. It is known that 1350 cm^{-1} peak corresponds to the disordered and structurally defected regions consisting of the sp^3 hybridization carbon. On the other hand, the 1587 cm^{-1} peak corresponds to the stretching of the C=C bond in graphitic carbon with sp^2 hybridization state, indicating that MMAO-CS coating is composed of carbon black. The XPS, XRD and Raman data confirm the formation of carbon soot on the surface.

3.2. Adhesion and wettability of coatings at different modification stages

The bonding strength between the substrate and MMAO-CS coating is one essential factor in this composite coating. To know the adhesion strength of the different coating systems, the scratch test is carried out, and coating adhesion performance is shown in Fig. S2. It can be observed that the adhesion strength of MMAO/AZ31 and MMAO-CS/AZ31 interface is 14.91, and 30.16 N, respectively, implying that the addition of CS layer onto MMAO coating has increased the adhesion strength in comparison with the pure MMAO coating.

The surface wettability is an important aspect for the corrosion inhibition in both seawater and marine atmospheric condition. With an extraordinary repellence effort to water phase, superhydrophobicity can prohibit the water film formation on the external surface. Meanwhile, in the bulky seawater phase, the air cushion entrapped in the matrix will also inhibit the invasion of the corrosive species. A typical superhydrophobic coating is mainly determined by two features, contact angle (CA) more than 150° and rolling angle less than 5° . To illustrate the general repellence capability to liquid, the droplets commonly in daily life including milk, blue ink, juice, green tea and water behave spherical shape on MMAO-CS coating surface, and the value of contact angles is ca. 160° for all liquid droplets, showing excellent superhydrophobicity (Fig. 4a). Interestingly, these liquid droplets also easily roll off from the MMAO-CS coating surface without leaving the residual liquid. As a representative, on a slope with a low tilting angle of 1.9

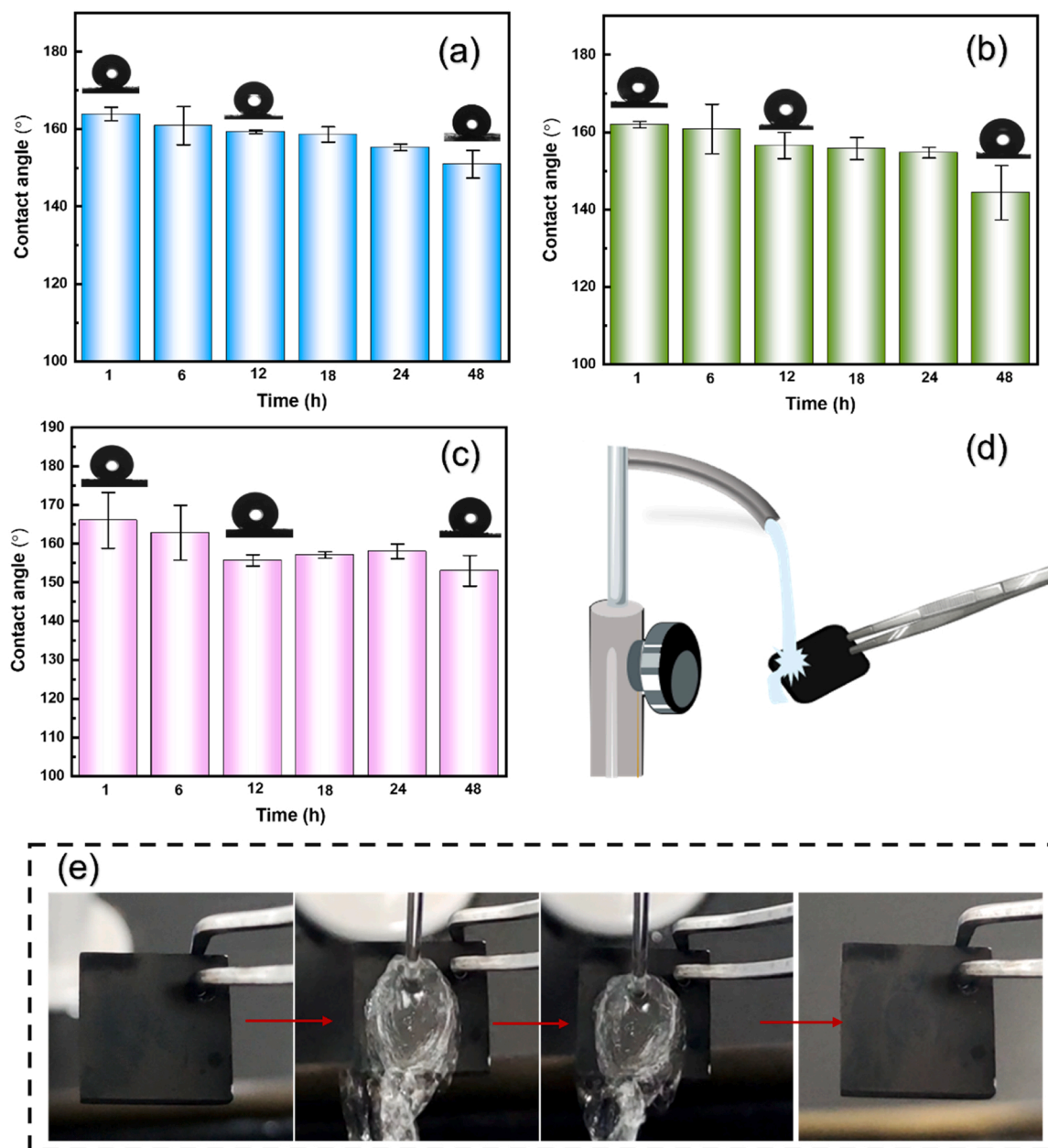


Fig. 6. Contact angle as indicating parameter for knowing the sustainability of MMAO-CS coating exposed to different solution (a) alkaline solution (pH=12), (b) acid solution (pH=3), and (c) nature seawater for different periods. (d) Schematic illustration of high-speed water jet for testing the sustainability of MMAO-CS coating. (e) The actual scenario of high-speed water jet to flush MMAO-CS coating.

$\pm 0.5^\circ$, a water droplet can easily roll off the MMAO-CS coating, revealing the excellent superhydrophobicity performance. It is suggested that MMAO-CS coating has very low affinity to the water related liquids.

Using the dropping-colliding-bouncing experiment, the dynamic wetting of the MMAO-CS is further evaluated. The water droplet will be released with different heights, including 1, 2, 5, and 10 cm. In Fig. 5, when a water droplet is released from 1 cm height upwards the target surface, the droplet can easily bounce off the surface after the collision, further demonstrating the MMAO-CS coating possessing excellent superhydrophobicity. Increasing the primary water droplet height is a facile way for upgrading the kinetic energy. When height of water drop is 2 cm, snapshots show the dropping-colliding-bouncing process on the MMAO-CS coating surface. The water droplet bounces on the surface before rolling off. No water is observed to reside on the surface when the

water droplet height adopts 1 and 2 cm. When the height continuously increases to 5 cm, the shape of water droplet extends to pancake style when reaching the lowest point, showing excellent superhydrophobicity. Furthermore, the kinetic energy of water droplet corresponding to height of 10 cm will see the different scenario. The falling water droplet breaks down to the small beads when contacting the SHS surface due to the higher kinetic energy afforded by the larger height. Then, the scattered water beads remain spherical shape on the surface to show the nonwetting phenomenon. After the high energy collision, the MMAO-CS coating can still maintain superhydrophobicity, illustrating the high robustness to the external environment.

Considering practical application of SHS coating in outdoor environment, the stability of MMAO-CS coating was evaluated by static immersion and water jet flush test. In Fig. 6a-c, for the initial immersion stage, the CA of alkaline (Fig. 6a) or acidic solution (Fig. 6b) on the MMAO-CS coating is basically in the range of $160\text{--}166^\circ$, demonstrating

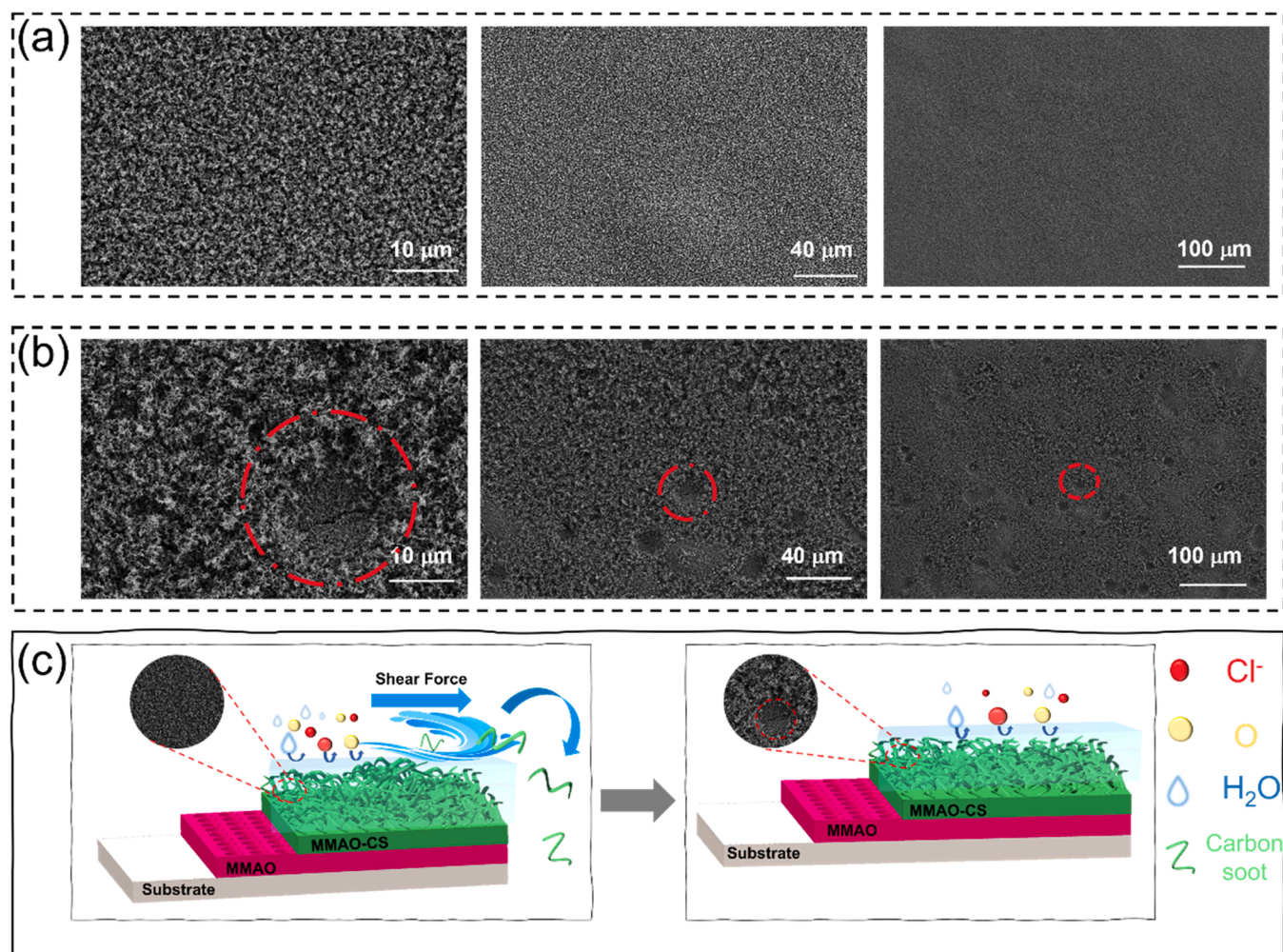


Fig. 7. Self-polishing of MMAO-CS coating. (a) The primary MMAO-CS coating without enduring the peeling off by shearing force from water. (b) The MMAO-CS coating after enduring the peeling off by immersion and shearing force from water. (c) Schematic self-polishing mechanism of as-prepared MMAO-CS coating.

excellent stability after the short immersion process. Even after 48 h, the SHS wettability can still maintain in the alkaline solution, which presents better sustainability in alkaline environment than in acid. From Fig. 6c, in the severely corrosive natural seawater environment, MMAO-CS coating surface still possesses good superhydrophobicity performance after 48 h. As shown in Fig. 6d, a simplified water jet test is carried out by controlling the water pressure of ca. 20 kPa with the incidence angle of 45° to the surface. Even after continuous flow flush for 1 min, the jet is completely repelled by the SHS without leaving trace of water residing on the surface, showing superior superhydrophobic property (Fig. 6e). In order to deeply understand the durability of MMAO-CS coating under the dynamic effect from water phase, the mechanism is analyzed in Fig. 7. SEM images show the direct morphological change of the surface before and after the immersion in water phase. Before the immersion, the primary MMAO-CS morphology is very even and flat, and no defect can be seen in the carbon soot matrix, proving the realization of an ideal film. The MMAO-CS sample was inserted into the acid solution for a certain period of immersion. After that, it was brought out from the solution. Therefore, it was endured two times of shearing action as well as the soaking from the water solution phase. From Fig. 7, it is seen that the attached loose CS layer partially detach from the top surface to behave a general “self-polishing” fashion, which is a very popular principle in the biofouling coating system. Moreover, for the sites occurring the self-polishing events, the underneath carbon soot still exists, and the SHS wettability still maintains.

Even if the external “floating” carbon soot is finally exhausted, the CS nanoparticles sticking and embedding in porous MMAO structure still maintain high surface roughness to show SHS wettability. Therefore, a soft carbon soot matrix affords an effective route to counter the short durability problem of the traditional SHS. During the peeling off of the external layer from carbon soot matrix, the failed SHS layer caused by vapor condensation and air dissolution will be polished off. Meanwhile, the inner fresh layer will expose to realize revival of SHS after self-polishing process. This versatile ideology is an important progress to break down the impasse that SHS is facing to.

3.3. Anti-fouling ability of self-polishing SHS

Biofouling is another sort of deterioration effect from the natural environment. When Mg alloy metal is proposed to be used in seawater, the biofouling has to be concerned, since the attached organisms in the natural seawater can accelerate Mg alloy corrosion. Previously, it is found that SHS is not so good as a candidate for the biofouling inhibition usage, because the dissolution of air cushion in the matrix will cause the rough structure to work as the anchoring sites for biofouling organisms. Now the chance for self-polishing SHS to be applied for biofouling is coming. The anchoring area can be flushed away by the shearing force from water to diminish the bio-attachment. Also, the carbon soot coating is composed by the discrete nanoparticles, which behave fluidity character. Thus, the modulus of the coating is very small, and the required

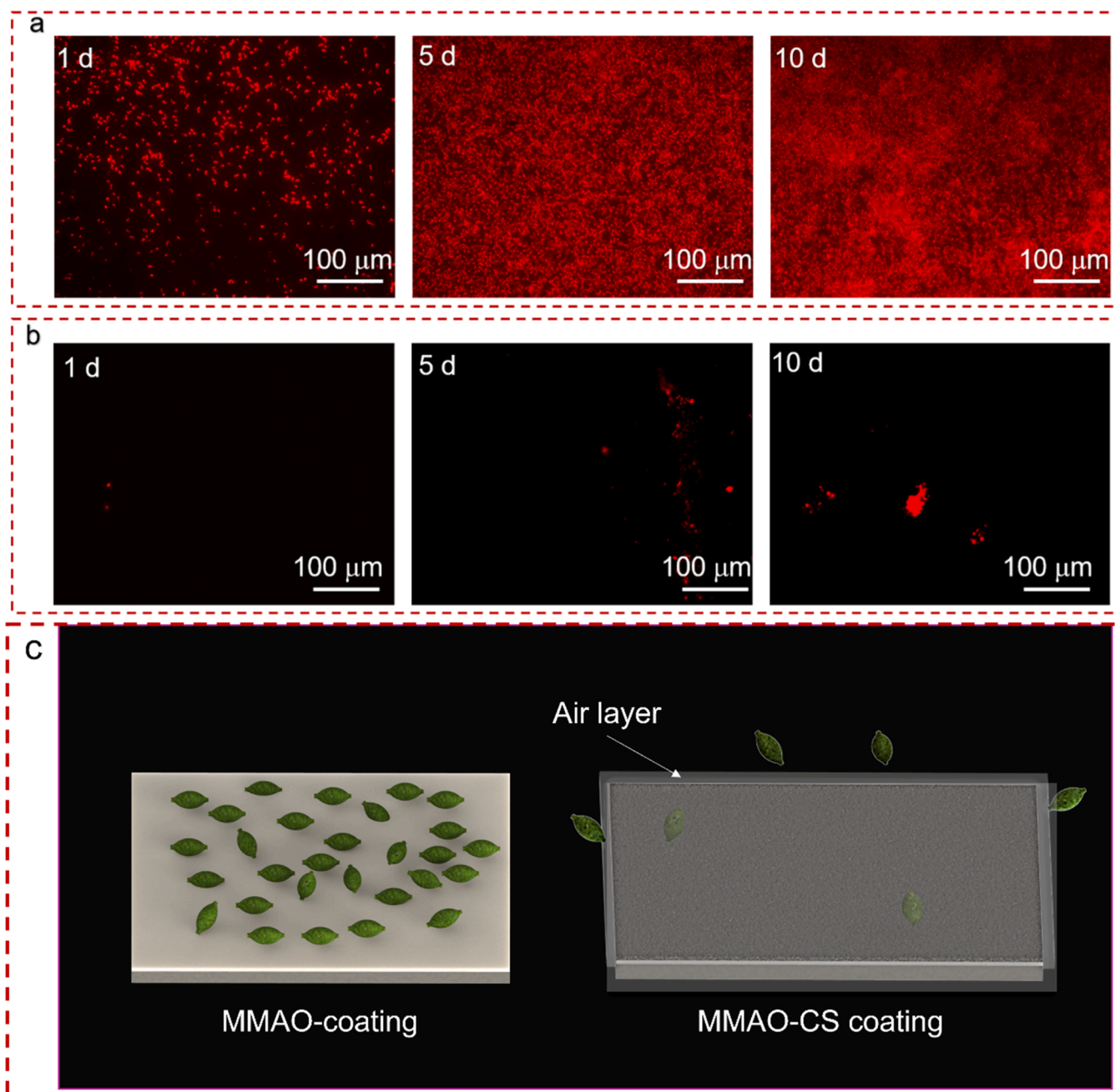


Fig. 8. Biofouling evaluation. Bio-attachment state of (a) MMAO and (b) MMAO-CS after immersion in diatom suspension for different time (1 d, 5 d and 10 d). (c) The relative bio-attachment mechanism for both MMAO and MMAO-CS coating after immersion in diatom suspension.

energy to detach the surface is very low. All these factors favor the biofouling inhibition. In order to explore the biofouling resistance of the MMAO-CS coating, diatoms are used as the target organisms in seawater environment. As shown in Fig. 8a, dense layer of diatoms covers the MMAO coating after immersion for 1, 5 and 10 d, respectively. For the first day immersion, few diatoms could be found to exist on the MMAO-CS coating (Fig. 8b). Even after immersion for 10 d, only a few diatoms can be detected to attach on the surface, showing good biofouling resistance in seawater.

3.4. Corrosion resistance of self-polishing SHS

To investigate the corrosion inhibition capability of the bare AZ31 alloy and different coatings, surface potential is monitored by using SKP

technique in the atmospheric condition. In Fig. 9, for a sample integrally containing bare AZ31 alloy, MMAO and MMAO-CS surface, it is obvious that the surface potential can be divided into three typical stripe regions. For bare AZ31 alloy, barrier layer for corrosion inhibition on the external surface is limited, and it is thermodynamically active with the potential distribution range from -1100 to -700 mV. In contrast, when AZ31 is covered by MMAO, the surface potential has remarkably changed to the range from -400 to -100 mV, indicating the corrosion susceptibility has been decreased. However, there are still many defects due to a large number of micropores in the matrix of MMAO. For MMAO-CS coating, the surface potential is from 690 to 866 mV. The super-hydrophobic coating prepared by the simple candle combustion method can fill the porous structure of MMAO and substantially improve the corrosion resistance to the underneath Mg alloy.

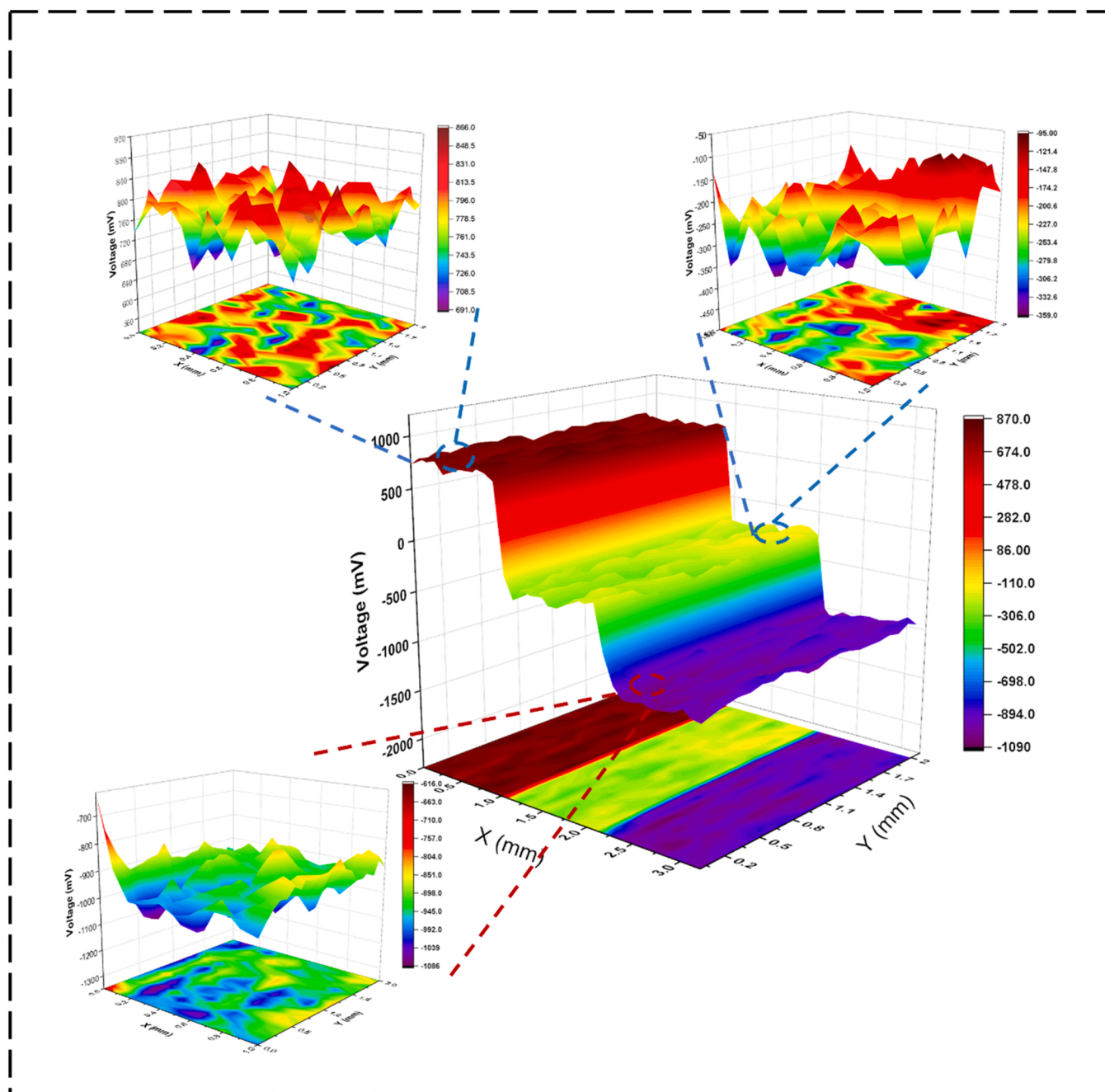


Fig. 9. SKP as the sophisticated tool to reveal surface potential distribution of the conjunction area covering bare AZ31 alloy/MMAO and MMAO/MMAO-CS. The up-left, up-right and down-left figures are corresponding to solo MMAO-CS, MMAO and bare AZ31 alloy.

EIS measurements are carried out to analyze the corrosion process and corrosion protection capability of the different coatings, and the results are shown in Fig. 10. In general, in the early stage of corrosion, the diameter of capacitance arc can directly reveal the protection efficiency from the as-constructed coating. For the bare AZ31, it shows the smallest capacitance loop, and this is attributed to the poor anticorrosion performance of the external oxide film. The semicircle diameter of MMAO-CS coating is much larger than that of bare AZ31 substrate and MMAO coating (Fig. 10a), demonstrating that MMAO-CS possesses better anticorrosion performance. From Fig. 10b, it is found that $|Z|_{0.01 \text{ Hz}}$ value of MMAO-CS coating is $2.82 \times 10^7 \Omega \text{ cm}^2$, which is much larger than that of MMAO coating ($6.19 \times 10^4 \Omega \text{ cm}^2$) and AZ31 substrate ($42.9 \Omega \text{ cm}^2$), respectively. On the other hand, in Fig. 10c, the phase angle is respectively -8.7° , -7.7° and 38.8° for AZ31 Mg alloy,

MMAO and MMAO-CS coatings at the low frequency of 0.01 Hz. And Bode diagram confirms that the phase angle of MMAO-CS coating shows a wider tendency than MMAO coating and bare substrate (Fig. 10c), implying that the carbon soot coating is efficient to protect the substrate from corrosion. The phase angle plot of AZ31 substrate has only one time constant, and the maximum phase angle is smaller than 90° , indicating there is a deviation from the ideal capacitance element. Fig. 10d–f shows the relative equivalent circuits (EC) corresponding to the different samples. The solution resistance and charge transfer resistance are represented by R_s and R_{ct} in the EC; the pitting corrosion event is entitled as R_L with an inductance L . For MMAO and MMAO-CS coating, the corrosion inhibition adopts complex series and parallel circuits indicated in Fig. 10e and f. In the EC of AZ31 substrate (Fig. 10d), at the high frequency, a capacitance loop is represented by R_{ct} and CPE_1 , which is

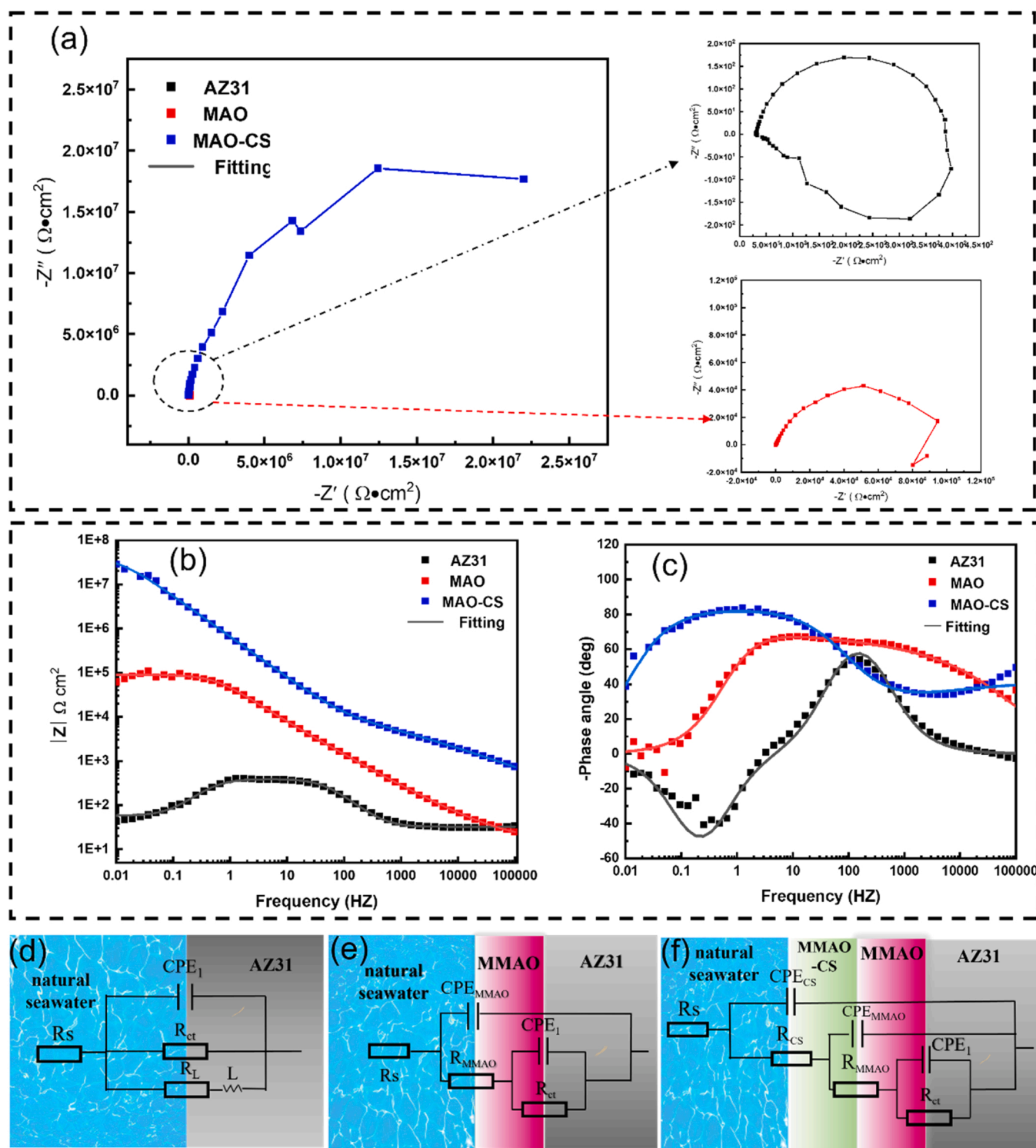


Fig. 10. Corrosion inhibition capability of bare AZ31 Mg alloy, MMAO and MMAO-CS revealed by EIS. (a) Nyquist plots, (b, c) Bode plots and equivalent circuit for (d) bare AZ31 Mg alloy, (e) MMAO and (f) MMAO-CS coating.

corresponding to the porous feature of corrosion product on the surface; the inductive loop appearing at the low frequency demonstrates the pitting corrosion begins to occur on the substrate surface (Fig. 10a). In the case of coating, the resistance and capacitance in the corrosion system will increase accordingly. For instance, in the EC of MMAO coating (Fig. 10e), R_{MMAO} describes the coating resistance. EC for MMAO-CS consists of R_{CS} and R_{MMAO} , respectively representing the outer CS coating and inner MMAO (Fig. 10f), which is different from

MMAO. Additionally, a larger R_{ct} value suggests higher corrosion resistance to the external environment. As listed in Table 2, MMAO-CS coating shows the largest R_{ct} value ($4.57 \times 10^7 \Omega \cdot \text{cm}^2$) compared with bare AZ31 substrate ($352.5 \Omega \cdot \text{cm}^2$) and MMAO coating ($8.40 \times 10^4 \Omega \cdot \text{cm}^2$), revealing that MMAO-CS coating has good corrosion resistance in seawater environment.

Table 2

Electrochemical data extracted from equivalent circuit fitting to the EIS curves.

Sample	R_s ($\Omega \bullet \text{cm}^2$)	CPE_1 ($\Omega^{-1} \bullet \text{cm}^{-2} \bullet \text{s}^n$)	n_1	R_{MMAO} ($\Omega \bullet \text{cm}^2$)	CPE_{CS} ($\Omega^{-1} \bullet \text{cm}^{-2} \bullet \text{s}^n$)	n_{CS}	R_{CS} ($\Omega \bullet \text{cm}^2$)	CPE_{MMAO} ($\Omega^{-1} \bullet \text{cm}^{-2} \bullet \text{s}^n$)	n_{MMAO}	R_{ct} ($\Omega \bullet \text{cm}^2$)	R_L ($\Omega \bullet \text{cm}^2$)	L
AZ31	32.38	1.04×10^{-7}	0.9192	—	—	—	—	—	—	352.5	27.4	106.5
MMAO	38	4.23×10^{-7}	0.7958	1334	—	—	—	1.55×10^{-6}	0.785	8.40×10^4	—	—
MMAO-CS	214.7	3.41×10^{-8}	0.8096	2659	8.21×10^{-8}	0.9157	9436	1.635×10^{-7}	0.9356	4.57×10^7	—	—

3.5. Long-term corrosion resistance of different coatings

As shown in Fig. 11, the natural seawater immersion test is carried out for evaluating the long-term corrosion resistance of the different coatings. The surface of AZ31 has been severely corroded after 24 h immersion, and the surface is covered with a layer of white corrosion products (the first two rows in Fig. 11). The SEM images of high magnification show that the corrosion product on the substrate surface is in spherical shape. However, there is no obvious change on the surface of MMAO and MMAO-CS, showing their good protection effect to the substrate after short-term immersion. Elongating the immersion time to 168 h, the AZ31 substrate color turns darker, and the corrosion product has densely distributed on the surface, which is caused by the intensified corrosion from the highly corrosive seawater. The SEM image shows that with the increase of soaking time, the corrosion product amount on AZ31 gradually increases and behaves a loose state, suggesting that a severe corrosion has occurred on the substrate. Therefore, it is revealed that the corrosion product on Mg alloy substrate cannot protect the underneath metal. Moreover, for MMAO, obvious macroscopic corrosion cannot be directly found after 168 h. However, making the comparison between the primary and as-immersed MMAO, the direct discrepancy can be observed. In Fig. S3, images shown in (a) and (b) illustrate that the porous surface is relatively flat, and no corrosion product exist on the primary MMAO surface. However, after immersion in seawater for 168 h, drastic surface change has been observed. The granular and irregular accumulation of corrosion product is formed on MMAO. Moreover, the magnified image in Fig. S3 (d) shows that irregular structure is growing from inside of the MMAO channel, which directly connects with the underneath Mg substrate. Therefore, the metal is assumed to be corroded to form the corrosion product growing out of the hole. The anti-corrosion effect will be weakened after long-term soaking. Unlike MMAO, MMAO-CS has no obvious change even after immersion for 168 h. The peeling off of CS in certain areas can be distinguished in SEM image. Since the loose CS nanoparticles existing at the external layer easily detach from the coating, the peeling off or the self-polishing will occur when the sample is endured the impact from the shearing force. However, no corrosion occurs on the surface of MMAO-CS due to the high water-repellency effect from SHS. Besides, from the perspective of wettability, the static contact angle of MMAO-CS remains at 156.75° after 168 h immersion, indicating that the coating still has good superhydrophobicity ability after long immersion process (Fig. S4). Therefore, MMAO-CS enables a tough barrier to block the corrosive medium, thus slowing down its entrance and extending the service life of the active substrate.

In summary, bare AZ31 alloy without protective coating will face severe corrosion from corrosive medium. After MAO treatment, the corrosion resistance of Mg substrate will be enhanced, since ceramic coating will act as the barrier to hurdle the occurrence of corrosion. However, MMAO coating intrinsically has numerous pores and channels in the ceramic layer, thus limiting the barrier effect. For MMAO-CS, it logically cuts the corrosion by the following strategies. Firstly, during the process for preparing MMAO-CS coating, the embedment of candle soot as well as the paraffin can effectively fill the porous structure of MMAO coating, so that the channels for corrosive species to go through is sealed, and the corrosion resistance is remarkably improved. Secondly, after the fabrication process, a composite superhydrophobic

MMAO-CS is formed. At the very external surface, a stable air cushion is formed with carbon soot as the rough micro-nano structure. The inner space of the bulky carbon soot matrix also traps a large quantity of air to form an air-solid composite coating. During the usage, the self-polishing will occur, and the new exposed carbon soot will also perform the SHS wettability. For the sites not having the self-polishing event, the entrapped air will be used to resist the penetration of the external corrosive species. Therefore, compared to bare AZ31 and MMAO, MMAO-CS has the most outstanding anticorrosion performance. In addition, the SHS based on CS affords an *opening coating system*, since the coating can be in situ fabricated as per necessity by using the facile combustion process. This is especially useful for the scenario of the periodic immersion corrosion condition, e.g., tidal area, the most severe corrosion zone in seawater environment.

4. Conclusions

In this work, a refreshable self-polishing superhydrophobic coating mainly composed of porous oxide and superhydrophobic carbon soot layer was prepared by MAO and candle soot deposition method. This MMAO-CS coating is proposed to be used for corrosion and biofouling protection of Mg alloy in marine environment. Wettability analysis shows MMAO-CS possesses excellent superhydrophobic property with a contact angle of 160° and prohibits the contamination from different aqueous media. The self-polishing and low modulus of the coating alleviates diatoms attachment to perform prominent biofouling inhibition. MMAO-CS endows high corrosion inhibition to Mg alloy, and $|Z|_{0.01 \text{ Hz}}$ value of MMAO-CS coating is 6 and 3 orders of magnitude larger than bare and MMAO covered Mg alloy substrate, demonstrating the promising role for corrosion protection. This proposed self-polishing superhydrophobic system based on CS can be in situ restored by using the facile combustion process, enabling a refreshable feature in harsh seawater environment.

CRedit authorship contribution statement

Bingzhi Li and Zhaoqi Zhang prepared the SHS materials. Also, the corrosion performance to the underlying metal in atmospheric environment was measured by him. The electrochemical test of the corrosion behavior in 3.5% NaCl solution or the bio-adhesion behavior of MMAO and MMAO-CS coatings mainly measured by them. The interpretation of the primary results was mainly conducted by them; Ri Qiu and Yibo Ouyang gave the general idea, wrote and revised the paper and afforded the instruments and financial support for this research; Li Wang and Yan Su performed XPS, XRD and FE-SEM characterization to the as-achieved materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

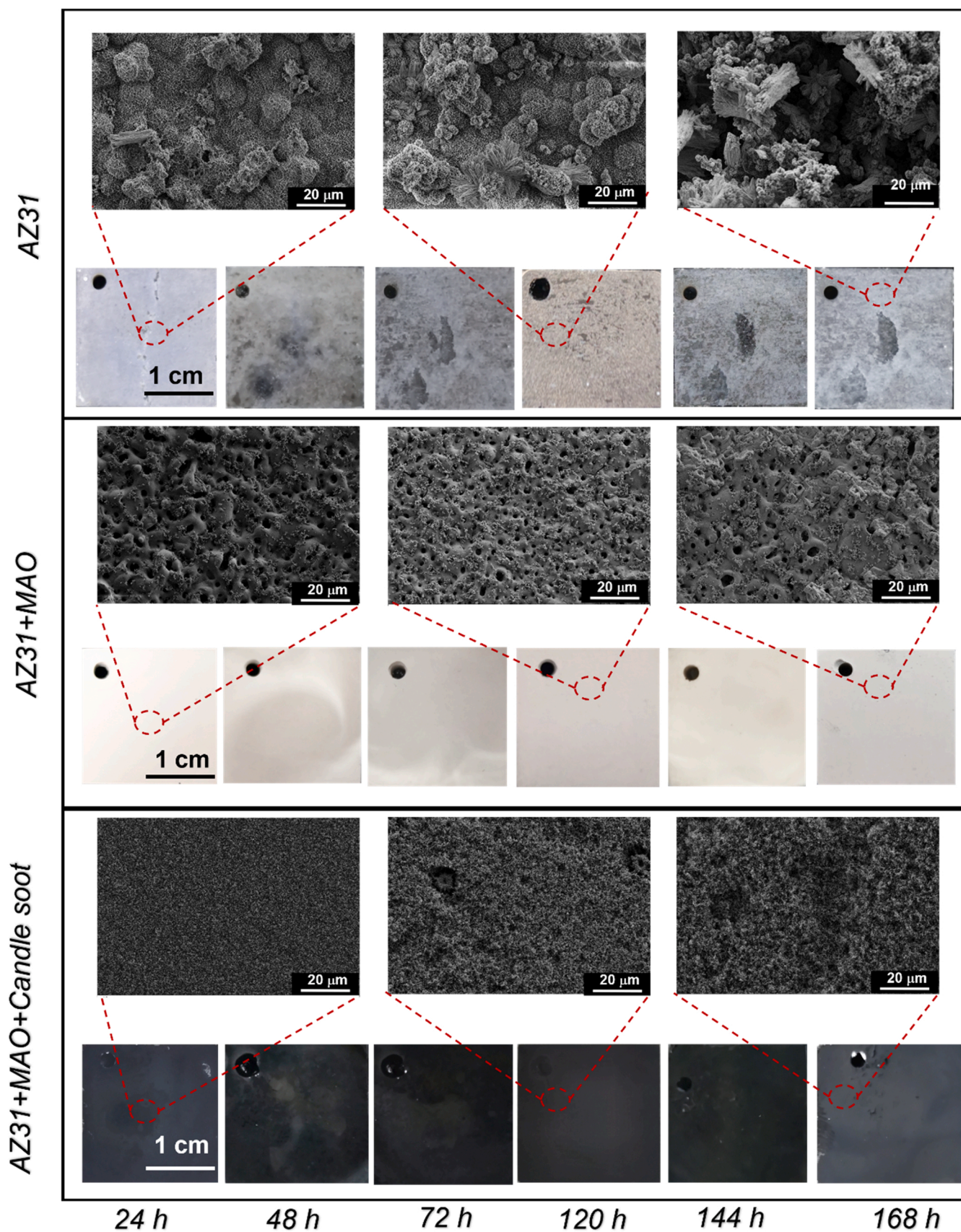


Fig. 11. Macroscopic and microscopic views of bare AZ31 alloy, MMAO and MMAO-CS after certain periodic immersion in natural seawater.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.colsurfa.2022.129693](https://doi.org/10.1016/j.colsurfa.2022.129693).

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